COMPLETED CHIRAL BAR-COBAR DUALITY AND MC4 WEIGHT CUTOFF

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Abstract. The classical bar-cobar adjunction of Priddy produces a Koszul dual for a quadratic algebra. For non-quadratic chiral algebras on a curve, completed bar constructions require two independent inputs: a degree cutoff for the Maurer-Cartan equation and a Mittag-Leffler condition for the inverse system of finite-stage cohomologies. A *strong completion tower* is a filtered augmented curved chiral A_∞-algebra whose operations are filtration-nondecreasing on the augmentation ideal. Under this axiom the Maurer-Cartan equation is componentwise finite. If the finite-stage cohomology tower is Mittag-Leffler, the completed bar-cobar comparison is a quasi-isomorphism in the square-zero total model; in the genuinely curved CDG ambient it is a coacyclic-cone comparison in the second-kind coderived category.

The standard conformal filtrations on affine Kac-Moody, Virasoro, principal $\mathcal{W}$-algebras, and lattice vertex algebras do not by themselves supply the algebra-level strong filtration axiom. They enter through bar-level weight-cutoff criteria. The positive MC4$^+$ regime is the $\rho = 0$ case, proved by strong completion or by a genuine auxiliary positive grading. The generic finite-resonance MC4^0 regime, including Virasoro and principal $\mathcal{W}_N$ at generic parameters, is proved on the positive-energy weight-completed finite-resonance surface. A bare screening presentation still requires the explicit chain-level strong deformation retract or an independent stabilization theorem that identifies it with that surface.

1. Introduction

1.1. The quadratic case and its limits. Classical Koszul duality is a phenomenon of graded algebras. A quadratic algebra $A = T(V)/(R)$ determines the dual coalgebra $A^{\text{i}} \hookrightarrow T^c(sV^*)$, the maximal subcoalgebra whose quadratic core is $s^2R^\perp$; the bar construction $B(A)$ mediates between them; and A is Koszul when $H^*(B(A))$ concentrates in bar length one, so that $\Omega(A^{\text{i}}) \simeq A$. The theory is due to Priddy [7], Beilinson–Ginzburg–Soergel [2], and in operadic form Loday-Vallette [4].

On an algebraic curve X, the objects of interest are chiral (vertex) algebras, whose relations are never quadratic in the strict sense. Heisenberg has a central term in the $J \cdot J$ OPE. Virasoro has a quartic pole. Principal $\mathcal{W}$-algebras $\mathcal{W}^k(\mathfrak{g}, f_{\text{prin}})$ have generators of weights $m_i + 1$, where m_i are the exponents of $\mathfrak{g}$; only for $\mathfrak{g} = \mathfrak{sl}_N$ are the weights $2, \dots, N$. Affine Yangians and quantum toroidal algebras have spectral-parameter relations of unbounded polynomial order. In each case the bar complex is a perfectly good dg coalgebra, but the iterated OPE produces bar elements whose differential contains infinitely many summands. A finite-dimensional truncation sees only a finite part of this object, and the classical bar-cobar adjunction is silent.

1.2. The completion problem. The remedy is completion. One introduces a decreasing filtration $\mathcal{A} = F^0\mathcal{A} \supset F^1\mathcal{A} \supset \cdots$ with finite-dimensional quotients $\mathcal{A}_{\le N} := \mathcal{A}/F^{N+1}\mathcal{A}$ and writes $\widehat{\overline{B}}(\mathcal{A}) := \varprojlim_N \overline{B}(\mathcal{A}_{\le N})$. At each finite stage N the strict bar-cobar adjunction produces a universal twisting morphism $\tau_N : \overline{B}(\mathcal{A}_{\le N}) \to \mathcal{A}_{\le N}$ which is a Maurer-Cartan element of the convolution dg Lie algebra $\mathrm{Conv}(\overline{B}(\mathcal{A}_{\le N}), \mathcal{A}_{\le N})$. The completion problem, labelled MC4 in [5], asks: do the finite-stage MC

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elements τ_N glue to a single MC element $\widehat{\tau}$ in the completed convolution algebra, and is the twisted tensor product acyclic in the limit?

Two obstructions appear. First, the MC equation $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$ is a sum over all bar degrees, and a priori each component is infinite. Second, the passage from finite-stage acyclicity to acyclicity in the limit requires the Mittag-Leffler condition, which is a separate finite-window hypothesis.

1.3. **The main result.** The central structural axiom is that the A_∞ operations of $\mathcal{A}$ respect the additive weight:

$$\mu_r(F^{i_1}\mathcal{A}, \dots, F^{i_r}\mathcal{A}) \subset F^{i_1+\cdots+i_r}\mathcal{A}. \tag{1.1}$$

A filtered curved chiral A_∞-algebra for which (1.1) holds and whose augmentation ideal $\overline{\mathcal{A}} = F^1\mathcal{A}$ is the first filtration piece is a *strong completion tower* (Definition 3.1).

Theorem 1.1 (Strong completion theorem, informal form)**.** *Let $\mathcal{A}$ be a strong completion tower whose finite-stage quotients lie in the proved square-zero bar-cobar regime and whose finite-stage cohomology tower is Mittag-Leffler. Then the completed bar construction $\widehat{\overline{B}}(\mathcal{A})$ is a separated complete pronilpotent curved dg chiral coalgebra, the completed convolution algebra carries a canonical Maurer-Cartan element $\widehat{\tau} = \varprojlim_N \tau_N$, and the counit*

$$\widehat{\epsilon}\colon \widehat{\Omega}\big(\widehat{\overline{B}}(\mathcal{A})\big) \xrightarrow{\ \sim\ } \mathcal{A}$$

is a quasi-isomorphism in the square-zero total model. In the curved CDG ambient the same finite-stage construction gives a coacyclic-cone comparison under the second-kind hypotheses.

The precise statement is Theorem 6.1 below. The proof uses three ingredients: (i) the degree cutoff lemma (§4), which reduces each component of the MC equation to a finite sum; (ii) the finite-window Mittag-Leffler criterion (§5); and (iii) the reduction principle of Lemma 5.2 which turns the finite-stage theory into a limit statement via the Milnor exact sequence.

1.4. **Positive versus resonant towers.** Completion problems split into positive and resonant regimes. A tower is *positive* if the filtration is defined by a genuine positive grading (e.g. the L_0-eigenvalue filtration on a graded vertex algebra). The degree cutoff lemma then holds after the strong axiom or the bar-level weight-cutoff criterion is verified. This is the MC4$^+$ regime.

A tower is *resonant* if there is a finite-dimensional weight-zero piece $R_{\mathcal{A}}$ on which higher operations can preserve filtration degree. The *resonance rank*

$$\rho(\mathcal{A}) := \dim H^*(R_{\mathcal{A}}, m_1^{R_{\mathcal{A}}})$$

classifies the completion difficulty. Positive towers have $\rho = 0$; Virasoro and the principal $\mathcal{W}_N$-algebras have $0 < \rho < \infty$; genuinely wild algebras have $\rho = \infty$. The MC4^0 problem concerns the middle regime. For Virasoro and principal $\mathcal{W}_N$ at generic parameters, the positive-energy weight-completed finite-resonance model proves the completed comparison. A concrete screening presentation of a principal $\mathcal{W}_N$ algebra enters this theorem only after one gives the chain-level strong deformation retract, or an independent weight-cutoff stabilization theorem, identifying it with that finite-resonance model.

1.5. **Strong filtration in curved bar-cobar theory.** Positselski [6] developed curved bar constructions for associative algebras with curvature. Loday-Vallette [4] supply the operadic bar-cobar machinery on which the chiral lift is built. The Beilinson-Drinfeld framework of filtered chiral algebras with I-adic topology [1] provides the ambient topology for the inverse limit. The theorem below is the chiral and A_∞ lift: the novelty is the strong filtration axiom, which identifies the precise structural hypothesis under which all finite-stage data assemble without further input.

2. The bar construction on the augmentation ideal

Throughout, X is a smooth complex algebraic curve and $\mathcal{A}$ is an augmented curved chiral A_∞-algebra on X in the sense of the monograph [5]. Augmentation means that $\mathcal{A}$ is equipped with a counit $\varepsilon\colon \mathcal{A} \to \mathbf{1}$ splitting the unit. The augmentation ideal is

$$\overline{\mathcal{A}} := \ker(\varepsilon).$$

Remark 2.1 (Augmentation ideal). The bar construction is built on $\overline{\mathcal{A}}$, not on $\mathcal{A}$: writing $T^c(s^{-1}\mathcal{A})$ in place of $T^c(s^{-1}\overline{\mathcal{A}})$ includes the unit and breaks the Koszul sign computation. We therefore set

$$\overline{B}^{\mathrm{ch}}(\mathcal{A}) := T^c\big(s^{-1}\overline{\mathcal{A}}\big),$$

with s^{-1} the homological desuspension $|s^{-1}v| = |v| - 1$ and with the chiral coproduct obtained by deconcatenation on the collision divisor of $\overline{C}_{\bullet+1}(X)$.

The curved chiral A_∞ operations $\{\mu_r\}_{r\geq 0}$ assemble into a codifferential $d_{\overline{B}}$ on $\overline{B}^{\mathrm{ch}}(\mathcal{A})$ with curvature coming from μ_0, so $\overline{B}^{\mathrm{ch}}(\mathcal{A})$ is a curved dg coalgebra. The convolution dg Lie algebra $\mathrm{Conv}(C, \mathcal{A}) := \mathrm{Hom}(C, \mathcal{A})$ has bracket from the coproduct on C and the product on $\mathcal{A}$; a Maurer-Cartan element is a degree-1 map $\tau\colon C \to \mathcal{A}$ with $\partial(\tau) + \tau \star \tau = 0$.

Theorem 2.2 (Finite-type bar-cobar duality, cited)**.** *Let $\mathcal{A}$ be a finite-type augmented curved chiral A_∞-algebra on X in the proved square-zero bar-cobar regime. Then the universal twisting morphism $\tau\colon \overline{B}^{\mathrm{ch}}(\mathcal{A}) \to \mathcal{A}$ is a Maurer-Cartan element of* $\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$*, and the counit*

$$\epsilon\colon \Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism in the square-zero total model. In the curved CDG ambient the corresponding finite-stage cone is coacyclic in the second-kind coderived category.

Proof. See [5, Thm. B]. The finite-type hypothesis reduces to finite-type vertex algebras, where the operadic bar-cobar machinery of Loday-Vallette [4] applies after the curved extension of Positselski [6]. □

3. The strong filtration axiom

Definition 3.1 (Strong completion tower)**.** An augmented curved chiral A_∞-algebra $\mathcal{A}$ on a curve X is a *strong completion tower* if it carries a descending filtration

$$\mathcal{A} = F^0\mathcal{A} \supset F^1\mathcal{A} \supset F^2\mathcal{A} \supset \cdots, \qquad \bigcap_{N\geq 0} F^{N+1}\mathcal{A} = 0,$$

such that:

(i) $\mathcal{A}$ is separated and complete: $\mathcal{A} \cong \varprojlim_N \mathcal{A}_{\leq N}$, where $\mathcal{A}_{\leq N} := \mathcal{A}/F^{N+1}\mathcal{A}$;
(ii) every quotient $\mathcal{A}_{\leq N}$ is of finite type and lies in the proved bar-cobar regime of Theorem 2.2;
(iii) $\overline{\mathcal{A}} = F^1\mathcal{A}$: the augmentation ideal is the first filtration piece;
(iv) the curvature term satisfies $\mu_0(1) \in F^1\mathcal{A}$, and all positive-arity chiral A_∞-operations are filtration-nondecreasing,
$$\mu_r(F^{i_1}\mathcal{A}, \ldots, F^{i_r}\mathcal{A}) \subset F^{i_1+\cdots+i_r}\mathcal{A} \tag{3.1}$$
for $r \geq 1$.

Axiom (3) ties the filtration to the augmentation. Axiom (4) is the structural heart of the definition: the curvature starts in the augmentation ideal, and the positive-arity operations respect the additive grading.

Remark 3.2 (On the quotient maps). Each projection $p_N : \mathcal{A}_{\le N+1} \twoheadrightarrow \mathcal{A}_{\le N}$ is a strict morphism of curved chiral A_∞-algebras, because the kernel $F^{N+1}\mathcal{A}/F^{N+2}\mathcal{A}$ is an A_∞-ideal of $\mathcal{A}_{\le N+1}$ by (3.1). The finite-stage bar constructions $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N})$ therefore form an inverse system of curved dg chiral coalgebras with strict morphisms, and the completed bar construction is the literal inverse limit.

Proposition 3.3 (Standard landscape weight cutoff)**.** *For the standard conformal-weight filtration $F^N\mathcal{A} := \bigoplus_{h\ge N}\mathcal{A}_h$, the following families satisfy the bar-level weight-cutoff criterion: in every fixed total conformal weight only finitely many generators and OPE residue terms contribute to the completed bar differential.*

(a) *the affine Kac-Moody vertex algebra $V_k(\mathfrak{g})$ for any simple $\mathfrak{g}$ and any non-critical level $k \neq -h^\vee$;*
(b) *the Virasoro vertex algebra Vir_c for any c;*
(c) *the principal $\mathcal{W}$-algebra $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ for any simple $\mathfrak{g}$ and any non-critical k;*
(d) *the lattice vertex algebra V_Λ for any positive-definite even lattice Λ.*

This statement does not assert the algebra-level strong filtration axiom $\mu_r(F^{i_1}, \dots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$.

Proof. Each family is L_0-graded with finite-dimensional homogeneous pieces and finitely many strong generators in each bounded weight window. The OPE of two homogeneous fields of weights a, b has only finitely many poles, and every residue term has conformal weight at most $a + b - 1$. Thus a bar chain of fixed total weight uses only fields of bounded weight and only finitely many residue terms. This is the bar-level cutoff needed for the finite-window MC4 comparison.

The assertion is weaker than algebra-level strong filtration. For affine currents, for example, the term $[J^a, J^b]$ has weight 1 although the inputs have weights 1 and 1; hence the strong axiom fails for the raw conformal filtration. Principal $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ is governed by the exponents of $\mathfrak{g}$: the generator weights are $m_i + 1$, with m_i running through the exponents. □

Example 3.4 (Non-example: $\beta\gamma$-system with conformal grading)**.** The $\beta\gamma$-system has a weight-$\frac{1}{2}$ grading that refines the conformal grading. With respect to the conformal-weight filtration alone, the strong filtration axiom fails because the OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ produces a weight-zero output from weight-$\frac{1}{2}$ inputs, violating additivity. Its shadow tower is contact class C, with a nonzero quartic contact shadow; it is not a Heisenberg-type degree-2 tower. A different auxiliary filtration may still give a bar-level weight cutoff.

4. The degree cutoff lemma

The structural consequence of the strong filtration axiom is that in the finite-stage bar complex, only finitely many degrees contribute. This is the precise mechanism by which the MC equation becomes componentwise finite.

Lemma 4.1 (Degree cutoff)**.** *Let $\mathcal{A}$ be a strong completion tower and let $\mathcal{A}_{\le N} = \mathcal{A}/F^{N+1}\mathcal{A}$ be the N-th quotient. In the finite-stage convolution algebra $\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N}), \mathcal{A}_{\le N})$, the Maurer-Cartan equation*

$$\partial(\tau_N) + \tau_N \star \tau_N = 0$$

involves only degrees $r \le N$: on $\mathcal{A}_{\le N}$ the bar differential is the finite sum $b_0 + b_1 + \cdots + b_N$ of coderivations coming from $\mu_0, \mu_1, \dots, \mu_N$.

Proof. The arity-zero curvature term b_0 is a single continuous term because $\mu_0(1) \in F^1\mathcal{A}$. Every positive-arity bar input lies in $\overline{\mathcal{A}}_{\le N}$, which by axiom (3) equals the image of $F^1\mathcal{A}$ in the quotient. The strong filtration axiom (3.1) gives

$$\mu_r(\overline{\mathcal{A}}^{\otimes r}) \subset \mu_r\big((F^1\mathcal{A})^{\otimes r}\big) \subset F^r\mathcal{A}.$$

Reducing modulo $F^{N+1}\mathcal{A}$, any term with $r \geq N+1$ lands in $F^r\mathcal{A} \subset F^{N+1}\mathcal{A}$ and vanishes in $\mathcal{A}_{\leq N}$. So the induced bar differential on $\mathcal{A}_{\leq N}$ is the finite sum $b_0 + b_1 + \cdots + b_N$, as claimed. □

Remark 4.2 (Degree cutoff as PBW degeneration). The degree cutoff is the operadic counterpart of the classical PBW theorem. For the universal enveloping algebra $U(\mathfrak{g})$ with its standard filtration by number of generators, the associated graded is $\mathrm{Sym}(\mathfrak{g})$; higher-degree brackets vanish modulo lower filtration stages. The strong filtration axiom extends this phenomenon to all chiral A_∞-operations on a graded vertex algebra: the degree cutoff lemma is the statement that at each finite stage, only finitely many Poisson (or vertex) operations are active. The classical PBW theorem is the $r = 2$ case.

Corollary 4.3 (Finite-stage MC assembly). *Let $\mathcal{A}$ be a strong completion tower. The universal twisting morphism τ_N at stage N is a Maurer-Cartan element of the finite-stage, locally finite convolution A_∞-algebra: the quotient $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ has finite-dimensional bar-length components, and the bar differential is the finite sum of Lemma 4.1.*

Proof. Finite type of $\mathcal{A}_{\leq N}$ gives finite dimension of each $\overline{\mathcal{A}}_{\leq N}^{\otimes r}$ for every r, and the degree cutoff lemma bounds the number of nonzero bar differential terms. Theorem 2.2 then gives the Maurer-Cartan property. □

5. Mittag-Leffler Finite-Window Criterion

The second obstruction to passing from finite-stage acyclicity to acyclicity in the limit is the possible failure of the Mittag-Leffler condition on the inverse system of finite-stage cohomologies. The strong filtration axiom supplies the degree cutoff; the Mittag-Leffler condition is a separate finite-window hypothesis.

Proposition 5.1 (Mittag-Leffler criterion for finite windows). *Let $\mathcal{A}$ be a filtered tower whose finite-stage bar complexes have finite weight windows: for each total weight w there is $N(w)$ such that the weight-w subcomplex of $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ is independent of N for $N \geq N(w)$. Then, for every cohomological degree m, the inverse system*

$$\left\{H^m(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))\right\}_{N\geq 0}$$

is Mittag-Leffler on every fixed weight slice. Consequently the finite weight truncations, and the product completion over weight slices, have vanishing $\varprojlim^1$.

Proof. The transition maps are surjections because the finite-stage bar construction is functorial on surjections of finite-type curved chiral A_∞-algebras. By hypothesis, for each fixed weight w there is $N(w)$ such that the transition map

$$\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1})_w \xrightarrow{\sim} \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})_w$$

is an isomorphism of subcomplexes for $N \geq N(w)$. Passing to cohomology gives stable isomorphisms on the weight-w slice.

For a finite weight truncation only finitely many slices occur, so the inverse system is Mittag-Leffler. For the completed product over all weights, inverse limits and products are taken slicewise over vector spaces; the preceding stabilization kills the derived limit on each slice, hence on the completed product. □

Lemma 5.2 (Complete filtered comparison). *Let $\{f_N : C_N \to D_N\}$ be a morphism of inverse systems of complexes with surjective transition maps. If each f_N is a quasi-isomorphism and the inverse systems $\{H^*(C_N)\}$ and $\{H^*(D_N)\}$ are Mittag-Leffler, then the induced map on limits $\widehat{f} : \widehat{C} \to \widehat{D}$ is a quasi-isomorphism.*

Proof. For any inverse system of cochain complexes the Milnor exact sequence reads

$$0 \to \varprojlim\nolimits^1_N H^{m-1}(C_N) \to H^m(\widehat{C}) \to \varprojlim_N H^m(C_N) \to 0.$$

Mittag-Leffler kills the derived limit term, and the map on honest limits is an isomorphism because each f_N is. □

6. The main theorem

Theorem 6.1 (Strong completion tower bar-cobar duality). *Let $\mathcal{A}$ be a strong completion tower (Definition 3.1) whose finite-stage comparison systems satisfy the finite-window Mittag-Leffler condition of Proposition 5.1. Write $\tau_N \in \mathrm{MC}\big(\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N}), \mathcal{A}_{\le N})\big)$ for the universal twisting morphism at stage N. Then:*

(i) Completed coalgebra. *The completed bar construction*

$$\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A}) := \varprojlim_N \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N})$$

exists as a separated complete pronilpotent curved dg chiral coalgebra with continuous differential, coproduct, and curvature. At each stage, $\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})/F^{N+1} \cong \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N})$.

(ii) Completed MC element. *The componentwise limit $\widehat{\tau} = \varprojlim_N \tau_N$ is a Maurer-Cartan element of the completed convolution algebra*

$$\widehat{\mathrm{Conv}} := \varprojlim_N \mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N}), \mathcal{A}_{\le N}),$$

and the MC equation $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$ converges in the inverse-limit topology, each component being a finite sum by the degree cutoff lemma.

(iii) Completed cobar. *The completed cobar object*

$$\widehat{\Omega}^{\mathrm{ch}}\big(\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})\big) := \varprojlim_N \Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N}))$$

is a separated complete curved dg chiral algebra, equal to the twisted tensor product $\mathcal{A} \otimes_{\widehat{\tau}} \widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})$.

(iv) Counit in the square-zero total model. *The completed counit*

$$\widehat{\epsilon}\colon \widehat{\Omega}^{\mathrm{ch}}\big(\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})\big) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism when both sides carry the square-zero total differential. In the genuinely curved CDG ambient, the same comparison is replaced by a coacyclic-cone statement in the specified second-kind coderived category.

(v) Dual comparison. *If $C = \varprojlim_N C_{\le N}$ is a separated complete pronilpotent curved dg chiral coalgebra whose finite quotients lie in the finite-type square-zero bar-cobar regime and whose finite-window tower is Mittag-Leffler, then the completed unit*

$$\widehat{\eta}\colon C \xrightarrow{\sim} \widehat{\overline{B}}^{\mathrm{ch}}\big(\widehat{\Omega}^{\mathrm{ch}}(C)\big)$$

is a quasi-isomorphism in the square-zero total model, with the curved CDG version read as a coacyclic-cone comparison.

(vi) Equivalence surface. *Completed bar and completed cobar give quasi-inverse equivalences on the homotopy category of strong completion towers satisfying the same square-zero and Mittag-Leffler hypotheses. In a curved CDG realization the equivalence is the corresponding second-kind coderived equivalence.*

Proof. Completed coalgebra. Each quotient map $p_N : \mathcal{A}_{\leq N+1} \twoheadrightarrow \mathcal{A}_{\leq N}$ is strict by axiom (4), inducing a strict morphism $q_N : \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1}) \to \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ compatible with differential, coproduct, coaugmentation, and curvature. The inverse limit exists componentwise with $b = \varprojlim b_N$, $\Delta = \varprojlim \Delta_N$, $h = \varprojlim h_N$; the curved coalgebra identities hold quotientwise and hence on the limit. Continuity is formal for the inverse-limit topology. Pronilpotence: the reduced coproduct on bar-length-$\leq m$ elements is nilpotent on each finite quotient, so the limit is topologically pronilpotent.

MC assembly. Define $\widehat{\tau} := \varprojlim_N \tau_N$. By Lemma 4.1, modulo $F^{N+1}\mathcal{A}$ only degrees $\leq N$ contribute to $\widehat{\tau} \star \widehat{\tau}$. The MC equation holds modulo F^{N+1} (where it reduces to the finite-stage equation for τ_N), hence on the completed coalgebra. This is claim (2).

Completed cobar. Each $\Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))$ is a curved dg chiral algebra; compatible transition maps give a complete curved dg chiral algebra in the limit, with continuous operations inherited from the finite stages. This is claim (3).

Counit. The quotient of $\widehat{\epsilon}$ modulo F^{N+1} is the finite-stage counit ϵ_N, which is a quasi-isomorphism by Theorem 2.2. The finite-window hypothesis gives the Mittag-Leffler condition for the source and target cohomology systems in Lemma 5.2. Applying that lemma to the compatible morphisms $\{\epsilon_N\}$ yields claim (4).

Dual comparison. The argument for $\widehat{\eta}$ is identical with bar and cobar exchanged, giving claim (5).

Equivalence. Claims (4) and (5) are the two triangle identities up to quasi-isomorphism, so completed bar and completed cobar form an adjoint equivalence on the homotopy categories restricted to strong completion towers. This is claim (6). □

7. Positive and Resonant Completion

The strong-completion theorem proves the formal MC4 mechanism once the degree cutoff, square-zero or coderived ambient, and Mittag-Leffler hypotheses are in place. The positive regime is the case $\rho(\mathcal{A}) = 0$: every completed operation either strictly raises the chosen positive filtration, or an auxiliary positive grading gives the same finite-window statement.

Corollary 7.1 (Positive MC4). *Let $\mathcal{A}$ be a positive strong completion tower with $\rho(\mathcal{A}) = 0$, or a tower equipped with a genuine auxiliary positive grading whose bar differential satisfies the finite-window cutoff and the square-zero or coderived ambient hypotheses of Theorem 6.1. Then the completed bar-cobar counit is a quasi-isomorphism in the square-zero total model, and a coacyclic-cone comparison in the curved CDG ambient.*

Proof. The strong case is Theorem 6.1. In the auxiliary positive grading case, positivity supplies the finite-window hypothesis of Proposition 5.1, so the same inverse-limit comparison applies. Finite-resonance towers with $0 < \rho(\mathcal{A}) < \infty$ are not included in this corollary. □

Definition 7.2 (Finite resonance). A filtered chiral algebra has finite resonance if the part of the bar differential preserving filtration degree is controlled by a finite-dimensional complex $R_{\mathcal{A}}$. Its resonance rank is

$$\rho(\mathcal{A}) = \dim H^*(R_{\mathcal{A}}).$$

The equality $\rho = 0$ is the positive case. The case $0 < \rho < \infty$ is a finite-dimensional obstruction problem requiring explicit homotopy data.

Theorem 7.3 (Generic finite-resonance MC4°). *Let $\mathcal{A}$ be Virasoro at generic central charge or a principal $\mathcal{W}_N$ algebra at generic shifted level, equipped with its positive-energy weight-completed finite-resonance model. Assume the resonance complex $R_{\mathcal{A}}$ is finite dimensional and is represented inside the completed bar complex by either a chain-level strong deformation retract or a finite-window stabilization. Then the MC4° completed bar-cobar comparison holds in the same square-zero, respectively coderived, ambient as Theorem 6.1.*

Proof. The positive-energy filtration decomposes the completed bar complex into finite weight windows. The part of the differential preserving filtration is, by hypothesis, represented by the finite complex $R_{\mathcal{A}}$; the remaining summands strictly raise weight and therefore satisfy the cutoff of Proposition 5.1. The chosen strong deformation retract, or equivalently the finite-window stabilization, identifies the finite-resonance summand with a fixed finite complex at every sufficiently large stage. Hence the inverse system of finite-stage cohomologies is Mittag-Leffler: the positive part stabilizes weight by weight, and the resonance part is finite dimensional. The completed comparison is then the inverse-limit comparison of Theorem 6.1. The coderived statement is obtained by replacing the square-zero quasi-isomorphism with the same finite-stage coacyclic-cone comparison before taking the Mittag-Leffler limit. □

Proposition 7.4 (Screening obstruction for the principal $\mathcal{W}_N$ route). *Let $\mathfrak{h}$ be the rank-$(N-1)$ Heisenberg vertex algebra with primitive chiral coproduct $\Delta_z^{\mathfrak{h}}$, and let Q_{α_i} be the Feigin–Frenkel screening charge* [3] *at shifted level Υ. A strong deformation retract presenting*

$$\mathcal{W}_N = \bigcap_i \ker Q_{\alpha_i} \subset \mathfrak{h}$$

must account for the chiral 1-cocycle

$$[Q_{\alpha_i}, \Delta_z^{\mathfrak{h}}] = \frac{\Upsilon - 1}{\Upsilon} \left(: e^{\alpha_i \cdot \varphi(0)} :\right) \otimes \left(: e^{\alpha_i \cdot \varphi(-z)} :\right) \cdot (1 + O(z)).$$

At $\Upsilon = 1$ this obstruction vanishes. At generic Υ it is nonzero and must vanish under the homotopy data or be absorbed in a completed coderived ambient.

Proof. This is the Vol. I screening-coproduct computation: expanding the primitive Heisenberg coproduct on the mode decomposition of $: e^{\alpha_i \cdot \varphi(\zeta)} :$ gives the one-loop contraction $\alpha_i^2 \log z = (2/\Upsilon) \log z$, whose regularized residue is $(\Upsilon - 1)/\Upsilon$. The same coefficient is the Miura cross-term coefficient for the spin-s principal $\mathcal{W}_N$ generators. □

8. Standard Families

8.1. $\beta\gamma$ and affine Kac-Moody.

For the $\beta\gamma_\lambda$ system the conformal weights are λ and $1 - \lambda$; the physical $\beta\gamma$ point $\lambda = 1$ gives weights 1 and 0. The raw conformal filtration therefore is not a positive strong completion tower. Its shadow data are contact class C:

$$S_3 = 0, \qquad S_4 = -\frac{5}{12}, \qquad S_r = 0 \quad (r \geq 5).$$

Any MC4 application must use an auxiliary filtration that proves the relevant bar-level cutoff.

For $V_k(\mathfrak{g})$ at non-critical level the conformal filtration gives the finite-window bar cutoff of Proposition 3.3. The algebra-level strong filtration fails because the $J^a_{(0)} J^b$ bracket has weight 1 from two weight-1 inputs. The critical level $k = -h^\vee$ lies outside the finite-stage proved regime because the Feigin–Frenkel centre changes the completion problem.

8.2. Virasoro and principal $\mathcal{W}$.

Virasoro has one weight-2 generator and OPE

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The central fourth-order pole is the finite resonance direction. Generic Virasoro MC4° is therefore the finite-resonance case of Theorem 7.3.

For principal $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ the generator weights are the exponents $m_i + 1$. In type A_{N-1} these are $2, \ldots, N$. The standard principal-stage W_∞ tower has a proved bar-level weight-cutoff route when the quotient system and square-zero/coderived hypotheses are specified. A generic Feigin–Frenkel screening descent for individual principal $\mathcal{W}_N$ stages must still confront Proposition 7.4.

Algebra	Input	Conclusion	Regime
$V_k(\mathfrak{g}), k \neq -h^\vee$	bar-level cutoff	proved with hypotheses	MC4$^+$
V_Λ	bar-level cutoff	proved with hypotheses	MC4$^+$
$\beta\gamma_\lambda$	auxiliary contact cutoff	requires cutoff	class C
Vir$_c$ generic	finite resonance model	proved with hypotheses	MC4°
principal $\mathcal{W}_N$ generic	finite resonance model	proved with hypotheses	MC4°
principal $\mathcal{W}_N$ screening presentation	screening SDR	requires SDR	descent
$V_{-h^\vee}(\mathfrak{g})$	Feigin–Frenkel centre	excluded	critical

References

[1] A. Beilinson and V. Drinfeld, *Chiral algebras*, American Mathematical Society Colloquium Publications, vol. 51, American Mathematical Society, Providence, RI, 2004.

[2] A. Beilinson, V. Ginzburg, and W. Soergel, *Koszul duality patterns in representation theory*, J. Amer. Math. Soc. **9** (1996), 473–527.

[3] B. Feigin and E. Frenkel, *Affine Kac–Moody algebras at the critical level and Gelfand–Dikii algebras*, Internat. J. Modern Phys. A **7** (1992), suppl. 1A, 197–215.

[4] J.-L. Loday and B. Vallette, *Algebraic operads*, Grundlehren der mathematischen Wissenschaften, vol. 346, Springer, Heidelberg, 2012.

[5] R. Lorgat, *Modular Koszul Duality, Volume I*, Monograph, 2026.

[6] L. Positselski, *Two kinds of derived categories, Koszul duality, and comodule-contramodule correspondence*, Mem. Amer. Math. Soc. **212** (2011), no. 996.

[7] S. B. Priddy, *Koszul resolutions*, Trans. Amer. Math. Soc. **152** (1970), 39–60.

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